

1) Develop a C program that demonstrates how a Linux operating system executes a command entered by a user that 1. Accept a Linux command as input. 2. Create a child process using fork(). 3. Execute the command in the child process using an appropriate exec() system call. 4. Allow the parent process to wait for the child using wait (). 5. Display the Process ID (PID) of both parent and child processes.

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>
#include <string.h>

int main()
{
    char cmd[100];

    printf("Enter Linux Command: ");
    scanf("%99s", cmd);

    pid_t pid = fork();

    if (pid < 0)
    {
        perror("Fork Failed");
        return 1;
    }
    else if (pid == 0)
    {
        printf("\n----- Child Process ----- \n");
        printf("Child PID : %d\n", getpid());
        printf("Parent PID : %d\n", getppid());

        execlp(cmd, cmd, NULL);

        perror("Command Execution Failed");
        exit(1);
    }
    else
    {
        printf("\n----- Parent Process ----- \n");
        printf("Parent PID : %d\n", getpid());
        printf("Child PID : %d\n", pid);

        wait(NULL);

        printf("\nChild Process Completed.\n");
    }

    return 0;
}
```

```
gowtham@DESKTOP-2F03K3T:~$ mkdir 2520090202_oss
gowtham@DESKTOP-2F03K3T:~$ cd 2520090202_oss
gowtham@DESKTOP-2F03K3T:~/2520090202_oss$ mkdir practical.c
gowtham@DESKTOP-2F03K3T:~/2520090202_oss$ cd practical.c
gowtham@DESKTOP-2F03K3T:~/2520090202_oss/practical.c$ mkdir practical-1.c
gowtham@DESKTOP-2F03K3T:~/2520090202_oss/practical.c$ cd practical-1.c
gowtham@DESKTOP-2F03K3T:~/2520090202_oss/practical.c/practical-1.c$ nano practical-1.c
gowtham@DESKTOP-2F03K3T:~/2520090202_oss/practical.c/practical-1.c$ gcc practical-1.c -o practical-1
gowtham@DESKTOP-2F03K3T:~/2520090202_oss/practical.c/practical-1.c$ ./practical-1
Enter Linux command: ls
```

Parent Process
Parent PID : 565

Child Process
Child PID : 566
practical-1 practical-1.c
Child Process Completed.
gowtham@DESKTOP-2F03K3T:~/2520090202_oss/practical.c/practical-1.c\$ cd ..
gowtham@DESKTOP-2F03K3T:~/2520090202_oss/practical.c\$